FINANCIAL RISK PLATFORM

WS5 | Backend Development Plan

AI/GenAI, Automation & Innovation Layer

Workstream objective: Add differentiated intelligence while keeping data access, financial calculations and consequential actions governed.

Product intent: A broad, lightweight, extensible platform foundation. Initial data volumes and module depth are limited deliberately; architecture, control points and expansion interfaces are not treated as throwaway demo code.

> **Synchronization rule: This document is paired with the opposite-side document for the same workstream. Feature IDs, phase IDs, endpoint contracts, statuses and acceptance criteria must remain aligned.**

Shared architecture: Layer 1 Universal Platform Foundation → Layer 2 Configurable Financial Domain → Layer 3 Financial Calculation Modules → Layer 4 Intelligence and Experience. Data governance and security cross-cut all layers.

1. Scope and Feature Coverage

Feature ID	Feature	Scope / implementation intent
AI-01	Natural-language rule proposals	Convert supported requests into structured, reviewable proposals, never arbitrary executable code.
AI-02	Research and analyst assistant	Retrieve from approved internal documentation, schemas, rule definitions and run evidence.
AI-03	Explainability assistant	Explain calculation differences using actual outputs, lineage and versioned rules.
AI-04	Agentic workflow orchestration	Invoke allowlisted APIs through scoped permissions, approval gates and logged tool calls.
AI-05	Document intelligence	Extract candidate fields from approved sample documents and route uncertain fields for review.
AI-06	Anomaly and drift intelligence	Suggest data anomalies or distribution changes, with confidence/evidence and human review.
AI-07	Regulatory/change knowledge	Track manually supplied or approved regulatory-change records; live feeds are optional future integration.
AI-08	AI governance and model controls	Prompt/data boundaries, model/version logging, evaluation, redaction and human oversight.

2. Phase-Wise Delivery

P0 | AI use-case and data boundary

Define permitted data, supported tasks, non-goals and human approval points.

P1 | Grounded assistant foundation

Retrieval over approved internal artefacts and metadata; show sources/evidence.

P2 | Structured proposals

Constrained rule proposals, schema/mapping suggestions and explicit review states.

P3 | Explainability and anomaly suggestions

Generate explanations from stored calculation evidence; anomaly suggestions require validation.

P4 | Controlled orchestration

Allowlisted service calls, scoped permissions, tool-call logs and approval before consequential actions.

P5 | Evaluation and governance

Test hallucination resistance, retrieval grounding, sensitive-data leakage and safe failure.

P6 | Optional integrations

Add document/OCR, external feeds or hosted models only when requirements, costs and security are approved.

3. Implementation Responsibilities

Backend responsibilities

• Implement a provider-neutral AI service interface; local/template mode must work without external APIs.

• Ground responses in approved retrieved artefacts and actual calculation/lineage data; return citations/evidence references where possible.

• Validate model output against strict schemas and allowlisted rules; never execute generated code.

• Enforce user permissions and data scope for retrieval and tool calls; redact or exclude sensitive data as policy requires.

• Log model/version, prompt template version, retrieval references, tool calls and approval decisions without leaking secrets.

• Build evaluation fixtures for groundedness, unsupported requests, data leakage and regression; treat live regulatory feeds and external screening as future integrations.

4. Paired API / Feature Contract Matrix

Endpoint names are proposed contracts for alignment, not a claim that the APIs already exist. Confirm payload schemas in the shared OpenAPI contract before implementation.

Shared feature pair	Capability	Proposed API contract
FE-AI-01 / BE-AI-01	Assistant request	POST /api/v1/ai/assist; GET /api/v1/ai/requests/{request_id}
FE-AI-02 / BE-AI-02	Rule proposal	POST /api/v1/rule-proposals; GET /api/v1/rule-proposals/{proposal_id}
FE-AI-03 / BE-AI-03	Explain run	POST /api/v1/ai/explanations; GET /api/v1/ai/explanations/{id}
FE-AI-04 / BE-AI-04	Controlled workflow	POST /api/v1/workflows; GET /api/v1/workflows/{workflow_id}; POST /api/v1/workflows/{id}/approval
FE-AI-05 / BE-AI-05	AI evaluation	POST /api/v1/ai/evaluations; GET /api/v1/ai/evaluations/{id}

5. Cross-Cutting Rules

• The canonical model and source mapping are versioned separately. Unknown source attributes are preserved or quarantined by explicit policy, not silently discarded.

• Every calculation run must bind dataset snapshot, schema/mapping version, rule-set version and calculation-engine version.

• Server-side authorization is authoritative. The frontend may hide unavailable actions but does not enforce security.

• No AI-generated code is executed. AI output is validated against schemas and policy, and consequential actions require authorized approval.

• All errors use a structured response with code, message, details and correlation_id. Sensitive payloads and secrets must not be logged.

• Prototype financial modules use synthetic/approved data, documented assumptions and test fixtures; do not claim regulatory certification or Moody's parity.

• The platform is designed for breadth, but production-grade depth is not implied for every financial domain. ALM/IRRBB, collateral, capital, live market data and regulatory feeds require their own specifications and validation.

6. Definition of Done

• Every feature ID in this document has a matching feature ID in its paired frontend/backend document and in the shared feature register.

• Each endpoint has request/response schemas, status enums, authorization expectations and error examples in OpenAPI.

• Frontend components use actual API responses or clearly labelled mocks matching the contract.

• Backend tests cover validation, permissions, version binding and domain logic appropriate to this workstream.

• The cross-workstream integration test passes for the features included in the release.

Change-control rule: Any change to endpoint, field name, enum, versioning behaviour or acceptance criteria must be reflected in both paired documents and the shared contract in the same change.